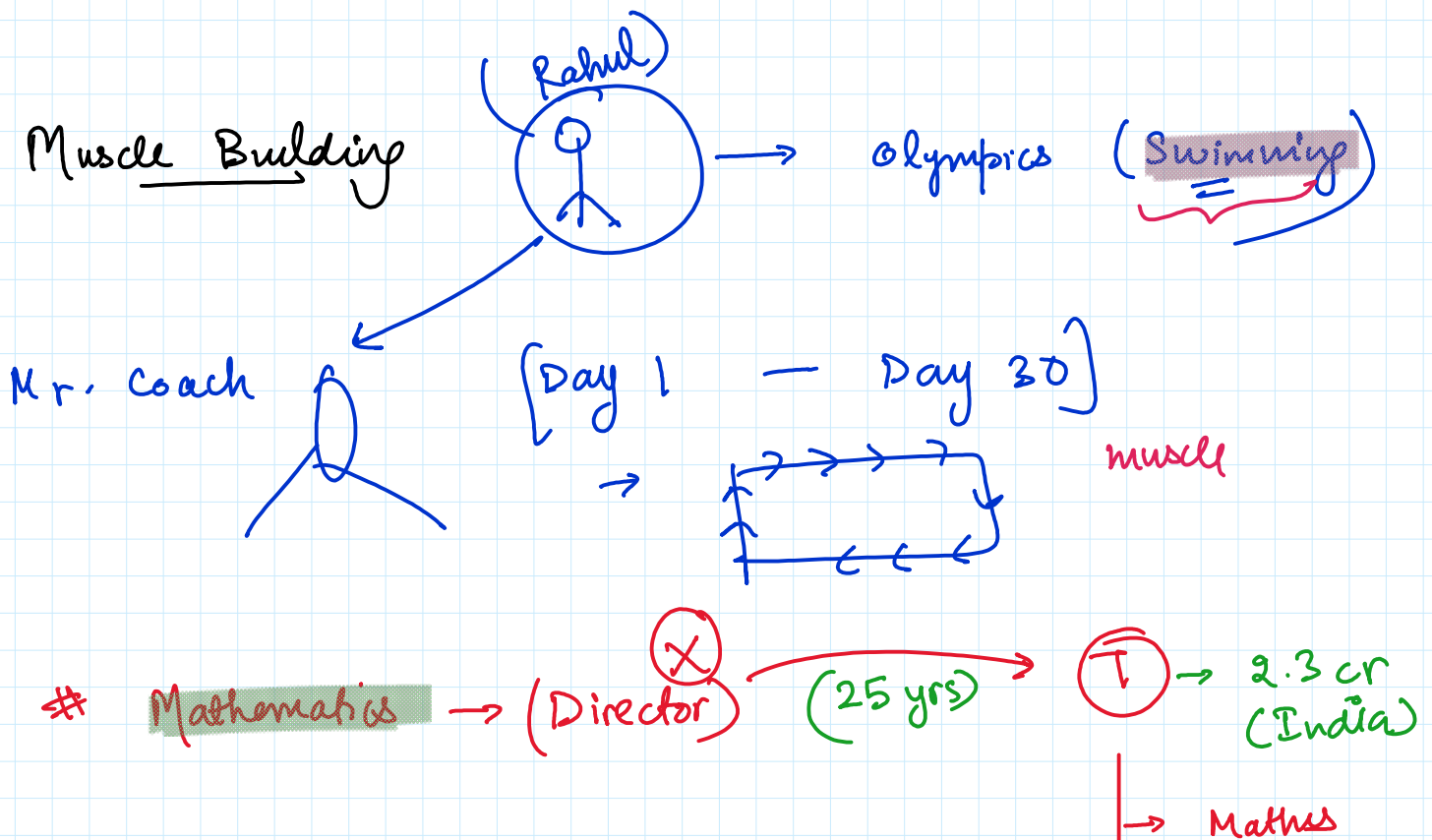


LA-4: (Loss minimisation in Classification) [+15 mins]

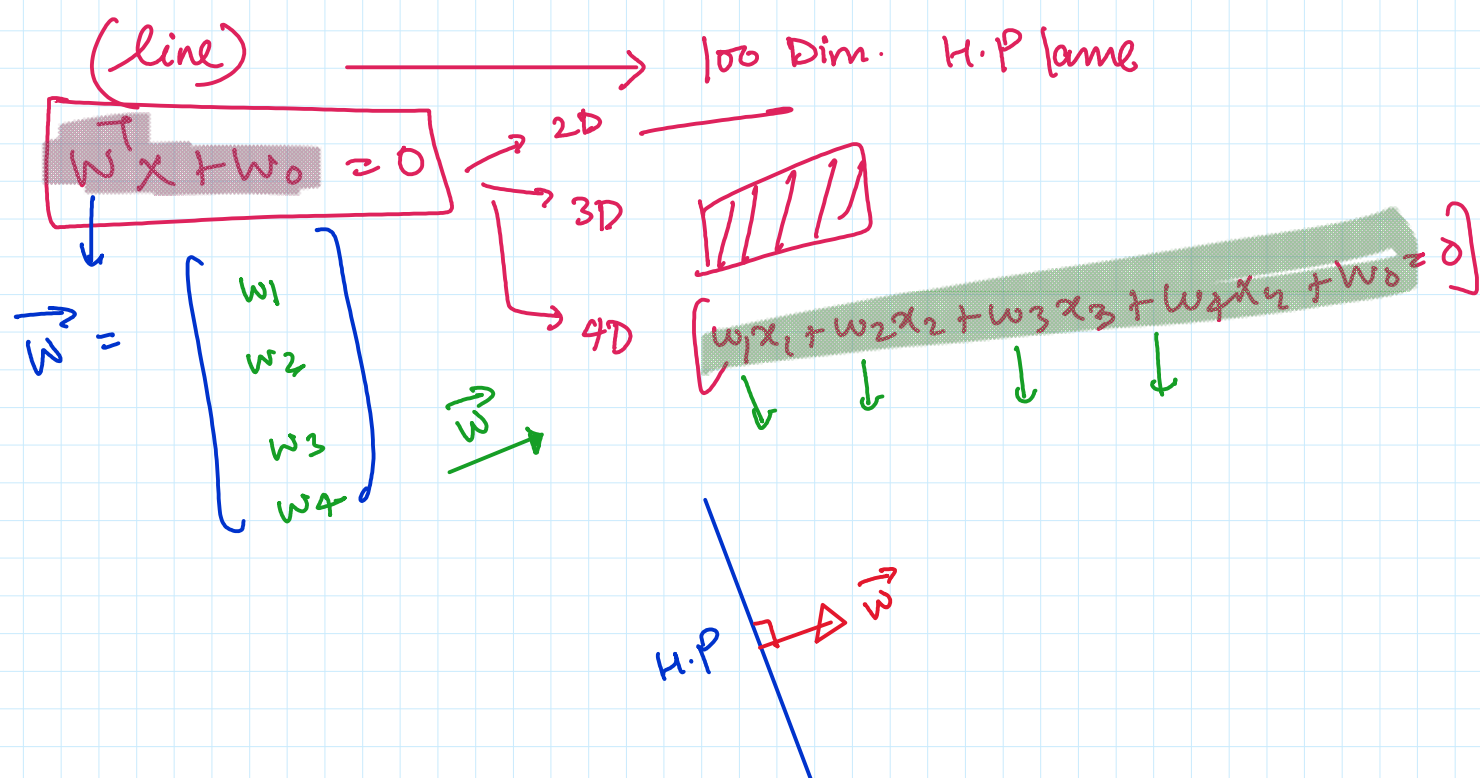
- Agenda → Introdⁿ
- Rohit Jindal
 - Sr. Data Scientist @ Tech comp
 - Masters IISc BLR (BA & DS)
 - Mastercard AI garage
 - Stall Street Gr.A
 - A.M @ IISc (GrOI)
 - Civil Engineer → D.C-E
 - University of Cambridge (EP & Sustainability)



→ Maths
→ Problem Solving

Recap why vectors → (Higher Dimension)

*(2D examples) → 100 Dimensions



Recap:

(a) Vectors: $\vec{x} \in \mathbb{R}^d$, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$

(b) Norm: $\|\vec{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2}$

(c) Inner Product: $\vec{x}, \vec{y} \in \mathbb{R}^d$
(Dot) $\vec{x} \cdot \vec{y} = \vec{x}^T \vec{y} = \sum_{i=1}^d x_i y_i$

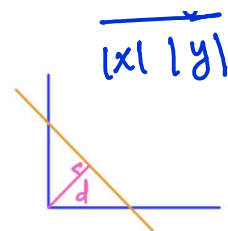
(d) Angle between vectors: $\cos \theta = \frac{\vec{x}^T \vec{y}}{\|\vec{x}\| \|\vec{y}\|} = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} \Rightarrow \frac{\|\vec{x} - \vec{y}\|^2}{\|\vec{x}\| \|\vec{y}\|} = \cos \theta$

(e) Distance b/w origin and line:

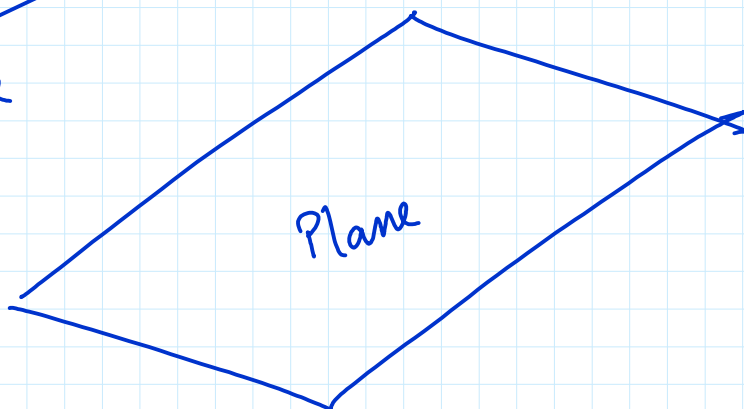
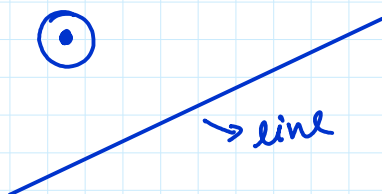
(c) Distance b/w origin and line:

line equation: $\vec{w}^T \vec{x} + w_0 = 0$

$d = \frac{w_0}{\|\vec{w}\|}$

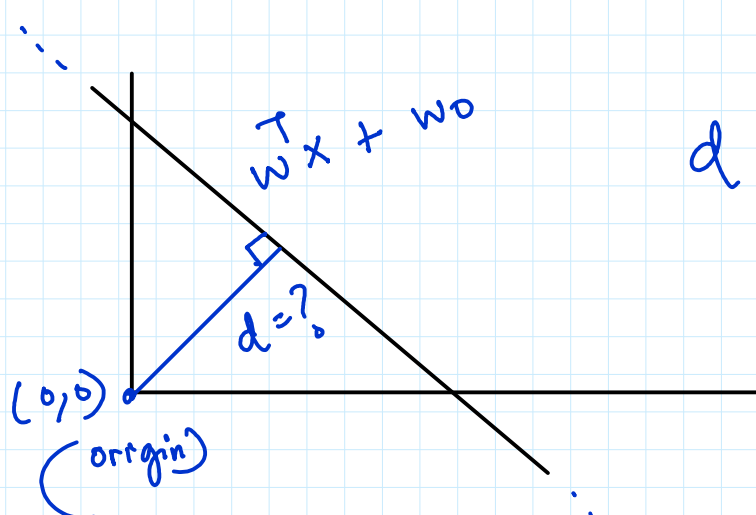


$|\vec{x}| |\vec{y}| \cos \theta$

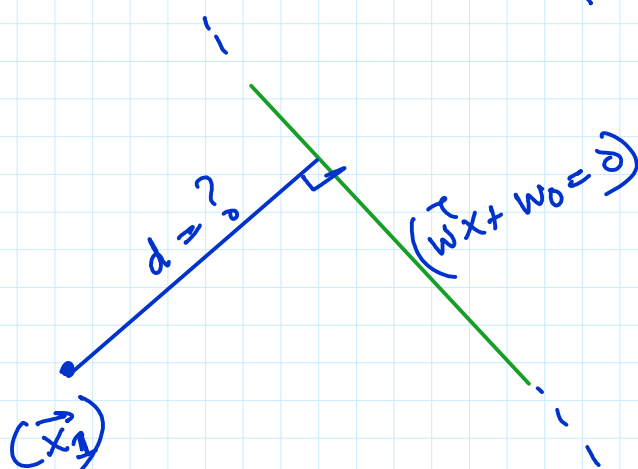


4D

5D



$d = \frac{\vec{w}^T \vec{0} + w_0}{\|\vec{w}\|}$



* (distance of H-P from any point)

$$d = \frac{|\vec{w}^T \vec{x}_1 + w_0|}{\|\vec{w}\|}$$

Ques

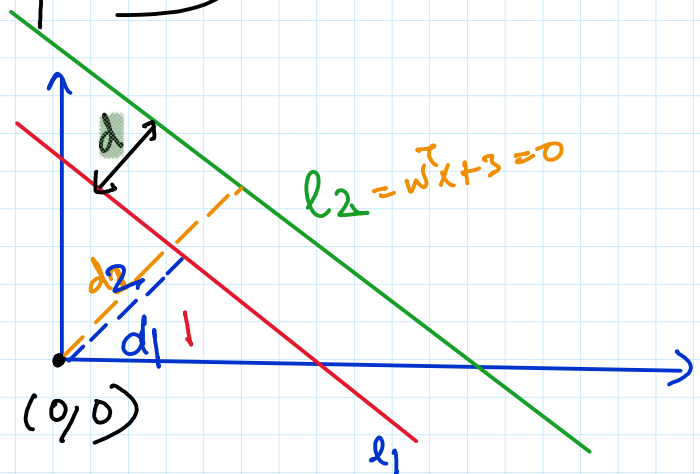
$$(2\underline{x_1} + 3\underline{x_2} + 4\underline{x_3} + 5\underline{x_4} - 7 = 0)$$

$$d = \frac{-7}{|\vec{w}| = \sqrt{2^2 + 3^2 + 4^2 + 5^2}} \quad \vec{w} = \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}$$

(Distance b/w 2 planes)

$$l_1 = \vec{w}^T \vec{x} + 5 = 0$$

$$l_2 = \vec{w}^T \vec{x} + 3 = 0$$



$$d = d_2 - d_1$$

$$\Rightarrow \left| \frac{3 - 5}{|w|} \right|$$

$$d_2 = \frac{3}{|w|}$$

$$d_1 = \frac{5}{|w|}$$

General formula
(distance b/w two hyperplanes)

$l_1 : \vec{w}^T \vec{x} + w_{01}$
 $l_2 : \vec{w}^T \vec{x} + w_{02}$

$$d = \frac{w_{01} - w_{02}}{|w|}$$

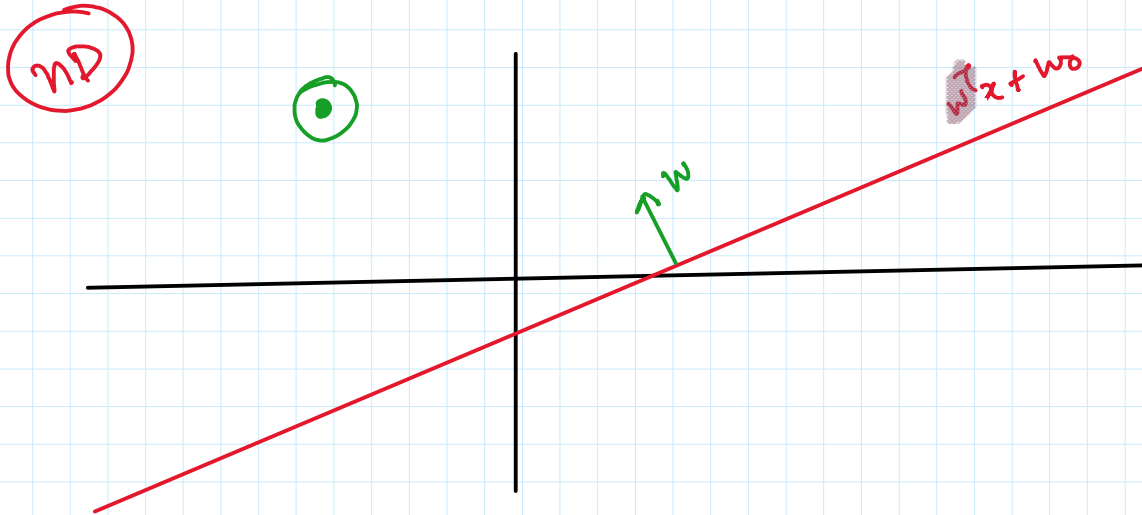
How to find the halfspace a point belongs to

(Eg.) $\Rightarrow \left(\underline{2x} + \underline{3y} - 4 = 0 \right) \rightarrow P \Rightarrow (2, 3)$

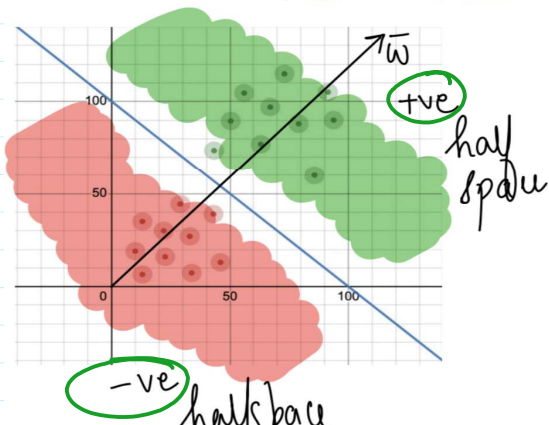
$$d \Rightarrow \frac{2(2) + 3(3) - 4}{|w|}$$

$$w = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

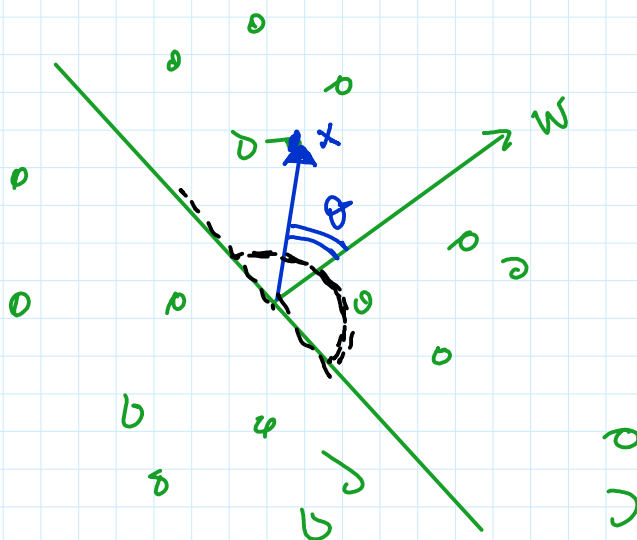
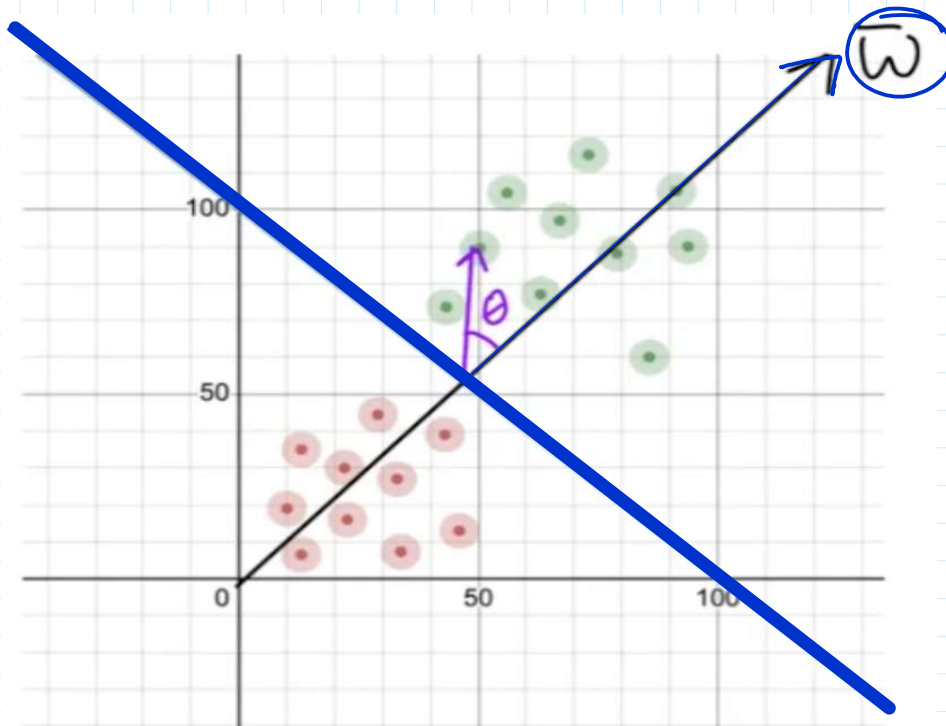
$$d = \frac{4 + 9 - 4}{\sqrt{2^2 + 3^2}} = \frac{9}{\sqrt{13}}$$



How to find the Halfspace a point belongs to

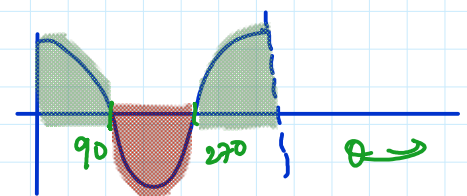


-ve halfspace



(angle b/w two vectors)

$\cos \theta$ *



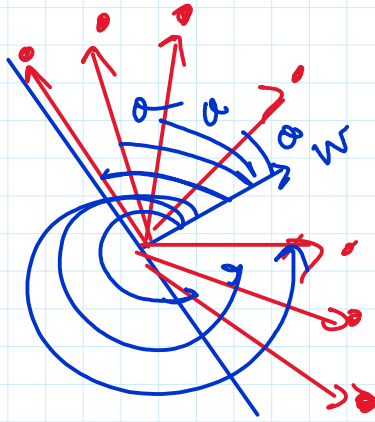
$\vec{w}$ and $\vec{x}$

$$\vec{w} \cdot \vec{x} = \overset{+}{|w|} \overset{+}{|x|} \cos \theta$$

sign ($\vec{w} \cdot \vec{x}$) is dependent on " θ "

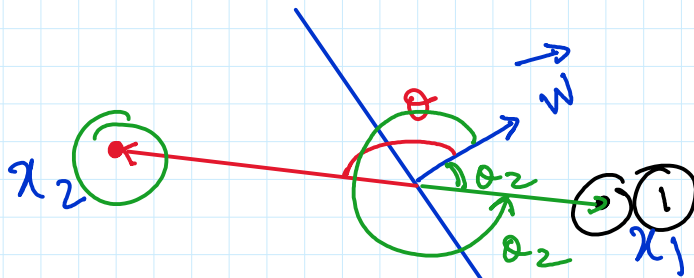
+ve $\leftarrow$ $\rightarrow$ -ve

+ve $\leftarrow$ $\rightarrow$ (-ve)

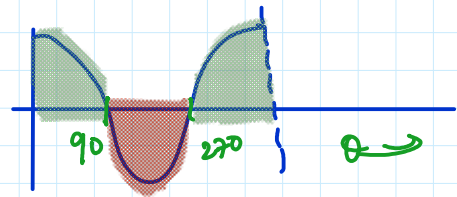


$(\theta \text{ b/w } 0 \text{ to } 90)$
 $(270 \text{ to } 360)$

$\vec{w} \cdot \vec{x}$ (+ve)



$\theta > 90^\circ \text{ and } < 180$



$$\cos \theta = -ve$$

$\Rightarrow (\vec{w} \cdot \vec{x} \text{ will also be } -ve)$

$$(\cos(360 - \theta) = \cos \theta)$$

$$\vec{w} \cdot \vec{x}_1 > 0$$

$$\vec{w} \cdot \vec{x}_2 < 0$$

$$\vec{w} \cdot \vec{x} = |\vec{w}| |\vec{x}| \cos \theta$$

Sign $(\vec{w} \cdot \vec{x})$ tells me the side

$$\vec{w} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$(2x + 3y = 9)$$

$$\vec{w} \cdot (2, 3, 1)$$

$$4x - 9y + 3z = 5$$

$$(0, 0, 0)$$

$$\text{sign}(\vec{w} \cdot \vec{x}) = \frac{4(2) - 9(3) + 3(1)}{\sqrt{4^2 + 9^2 + 3^2}} = \frac{-5}{\sqrt{130}}$$

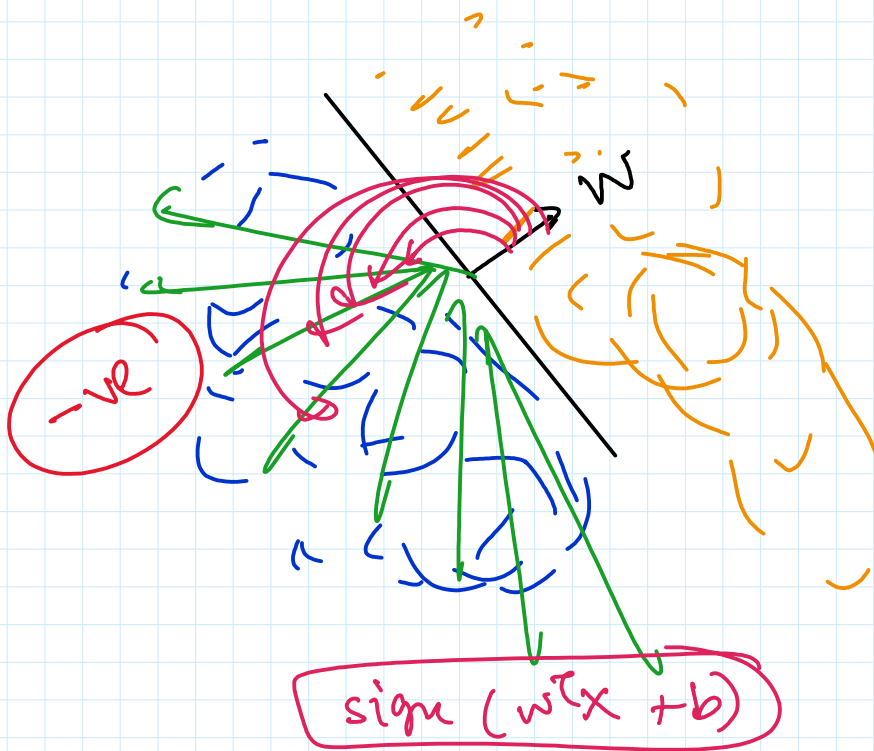
Dogs
(-1)

Cats
(1)

$\vec{w}$

* (cat / Dog)





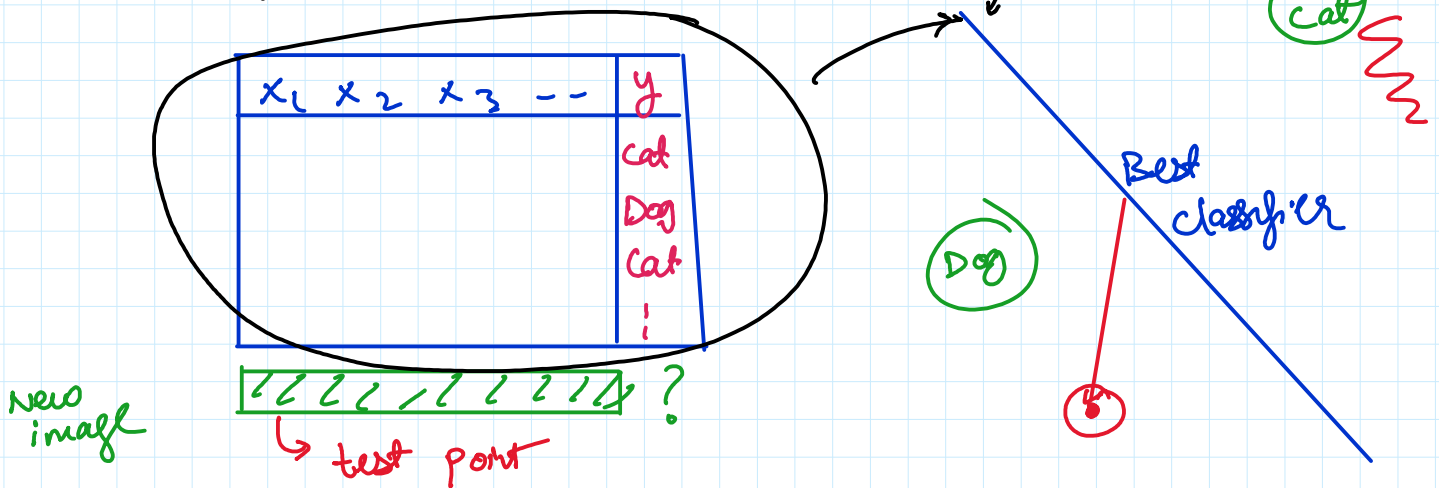
$$\vec{w} \cdot \vec{x} = |w| |x| \cos \theta$$

$$\vec{w} \cdot \vec{x} = |w| |x| \cos \theta$$

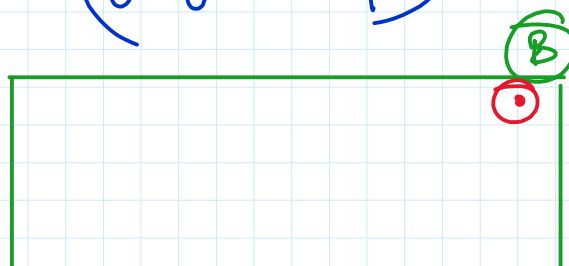
$$\text{sign}(w^T x + b)$$

$$\text{sign}(w^T x)$$

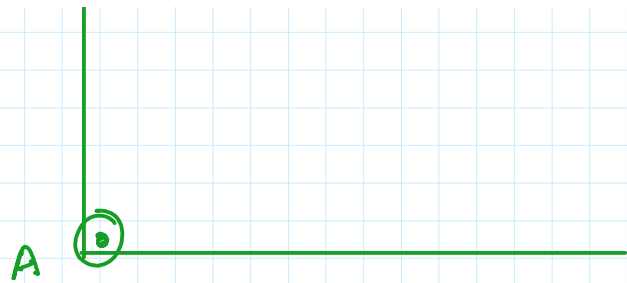
Loss function



Loss function $\rightarrow$ (google maps)



① (Take me in the shortest path)

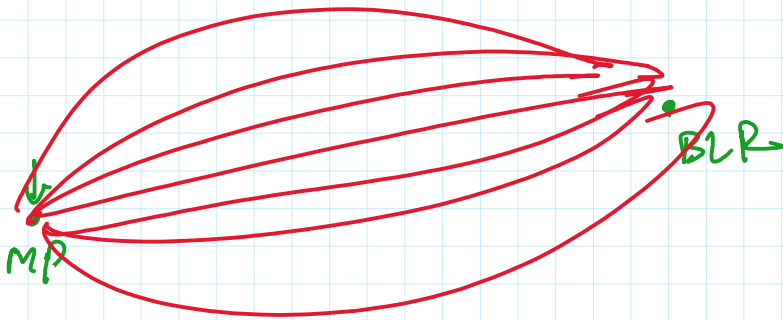


(shortest path)

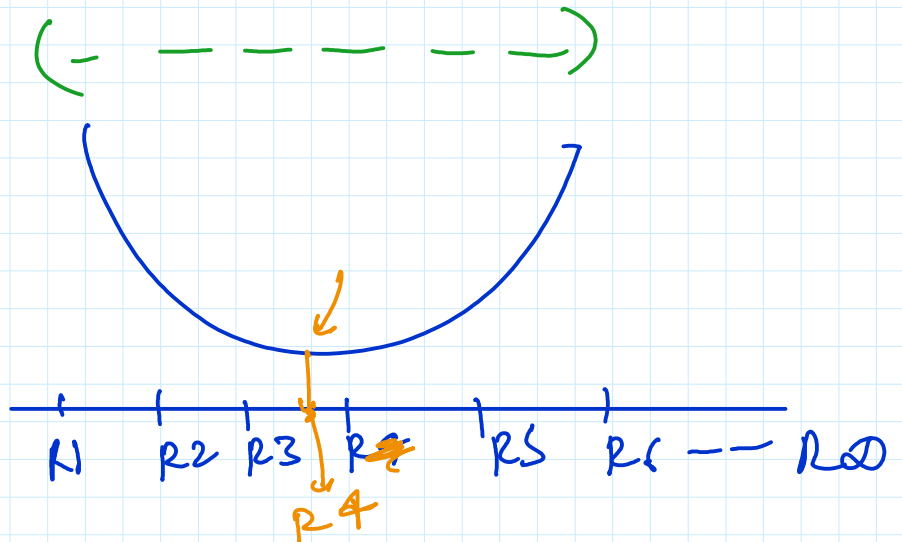


Mathematics

(Loss function)



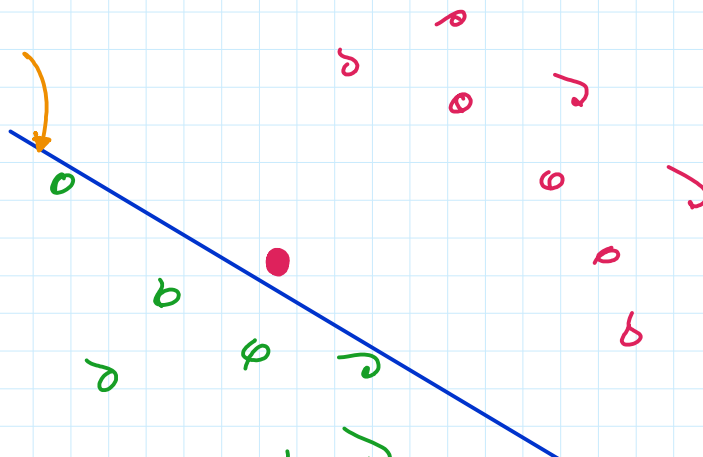
Loss function $\Rightarrow$

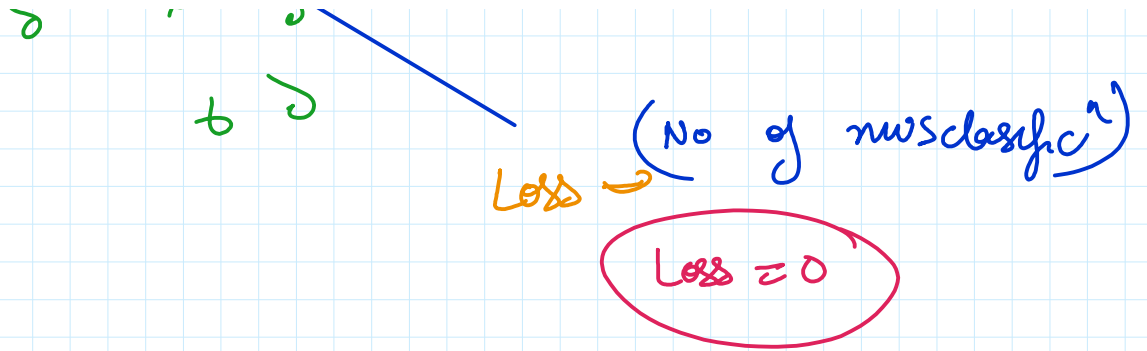


Ⓢ (Loss)

where my loss is minimum

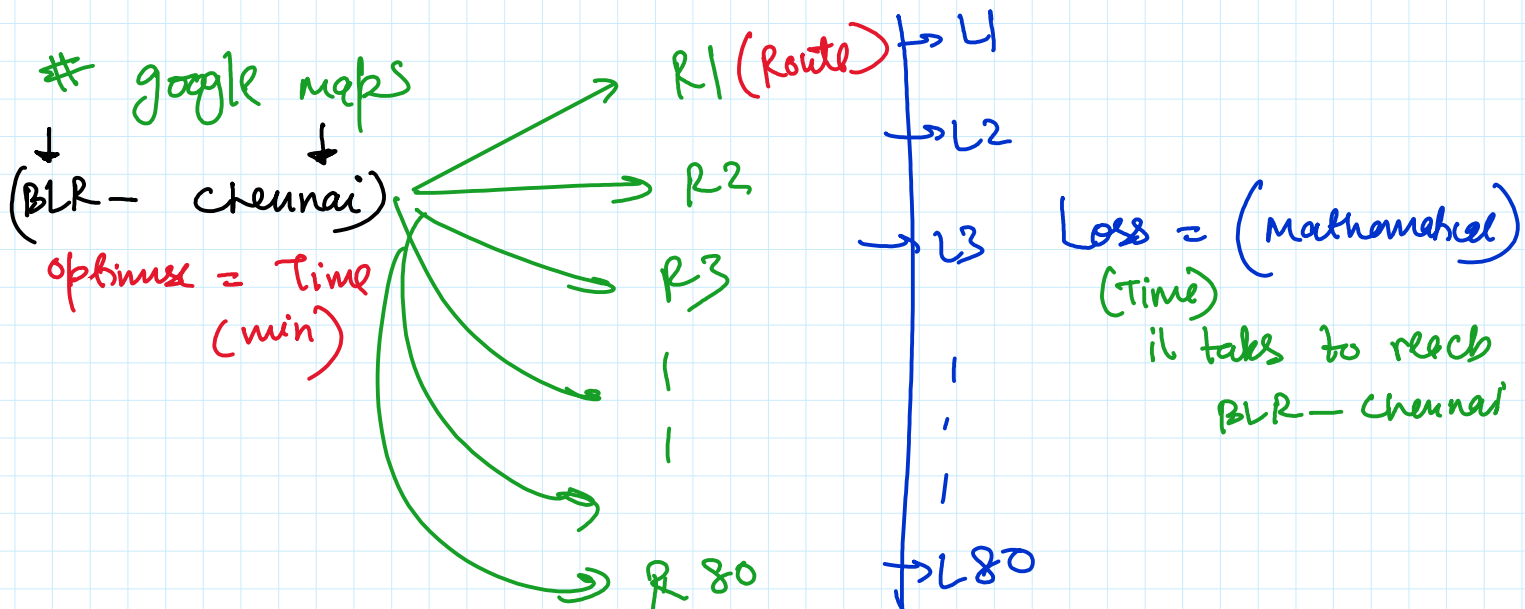
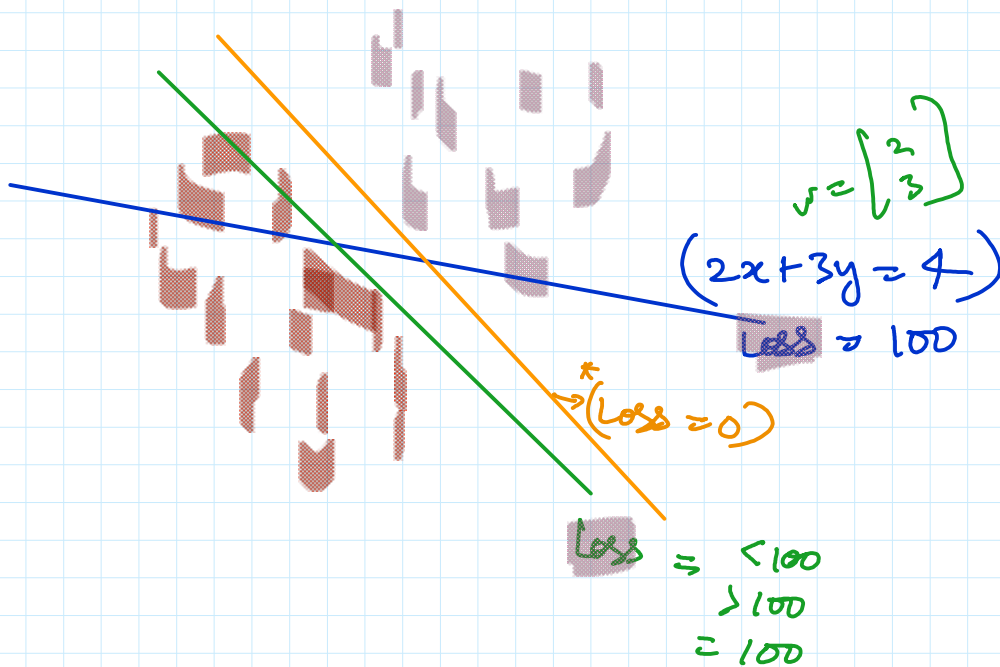
①





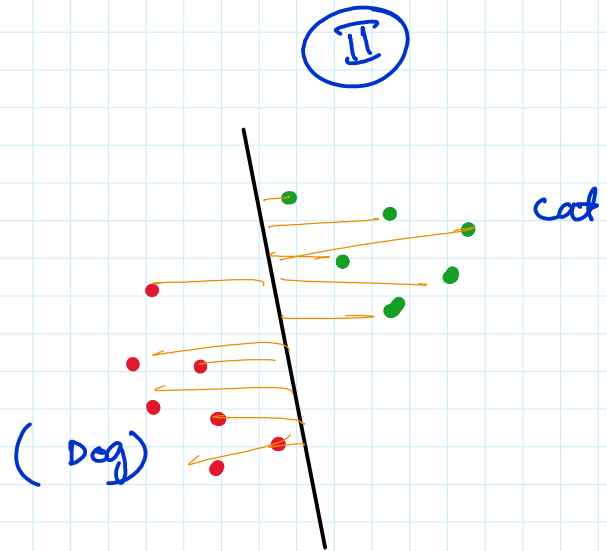
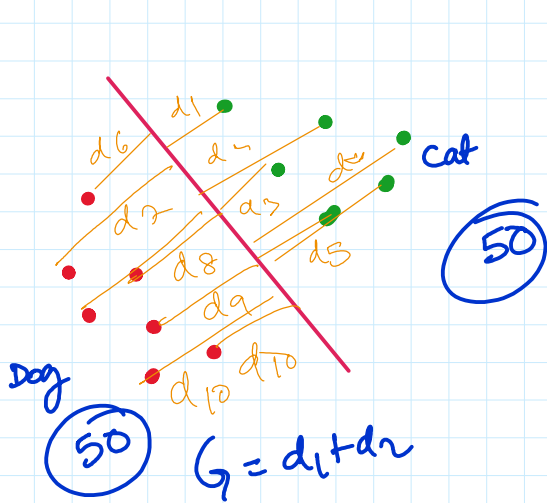
(5 min Break) = 10:17 pm *

(some more comments)



→ R 80 → L 80

(Sample loss function)



If we want to find the best classifier

the summation of the distances of the points from the hyperplane should be maximum

Gain function $\Rightarrow (d_1 + d_2 + d_3 + d_4 + \dots + d_{10})$

Best classifier $\Rightarrow$ (which max the gain function)

$$X = \{x_i y_i\}_{i=1}^n$$

$\swarrow$ $\searrow$
 $+1$ -1

x_1	x_2	y
		$+1$
		-1

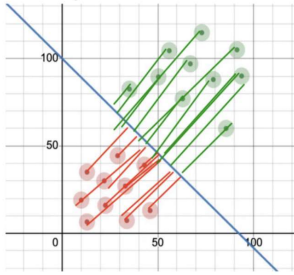
$w^T x + w_0 = 0 \rightarrow$ general classifier

$\cdot x_i$

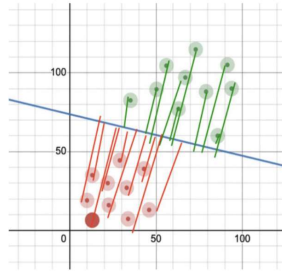
$(x \cdot w + w_0) = \sum_{i=1}^n w^T x_i + w_0 \rightarrow +ve, -ve$

$$G(X, \vec{w}, w_0) = \sum_{i=1} \frac{w^T x_i + w_0}{||w||} \rightarrow -\infty$$

Q) Which line in the figure below is a better comparison?



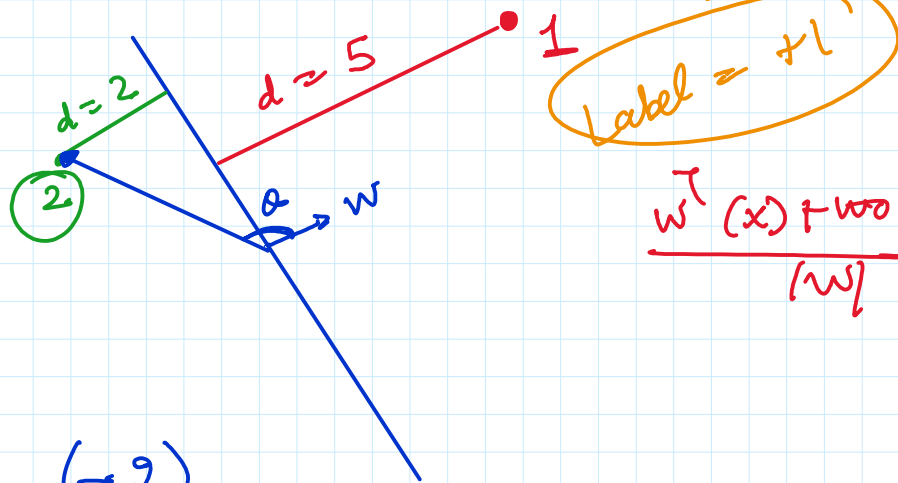
x_1, x_2	Act
●	-1
●	+1



We take distance
b/w feature vectors
& line

→ We want to maximise
the distance (Sum) b/w
feature vector & line.

Label = -1



$$\frac{w^T(x) + w_0}{||w||} = 5$$

$$\frac{w^T x + w_0}{||w||} = (-2)$$

$$G = \sum_{i=1}^n y_i \left(\frac{w^T x_i + w_0}{||w||} \right)$$

① Take mod (x)
② Square (x)

$$G = \sum_{i=1}^n y_i \left(\frac{\bar{w}^T x_i}{\|\bar{w}\|} \right) \rightarrow \textcircled{3}$$

x_1	x_2	y
		+1
		-1

$$+ \quad + \\ (-ve) \quad (-ve) = +ve$$

n/p. where (Label = 0, -1, same)

In order to find out best classifier, we must solve

$$\Rightarrow \text{MAX}_{\bar{w}, w_0} G(x, \bar{w}, w_0)$$

$$\Rightarrow \max_{\bar{w}, w_0} \left(\sum_{i=1}^n \left(\frac{\bar{w}^T x_i + w_0}{\|\bar{w}\|} \right) y_i \right)$$

$$\min_{\bar{w}, w_0} \left(- \sum_{i=1}^n \left(\frac{\bar{w}^T x_i + w_0}{\|\bar{w}\|} \right) y_i \right)$$

Loss function $\rightarrow$ (- Gain function)

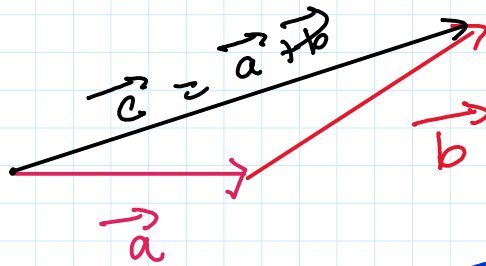
(max) $\xrightarrow{(-)}$ (minimization)

$$\max(x^2) \quad \approx \quad \min(-x^2)$$

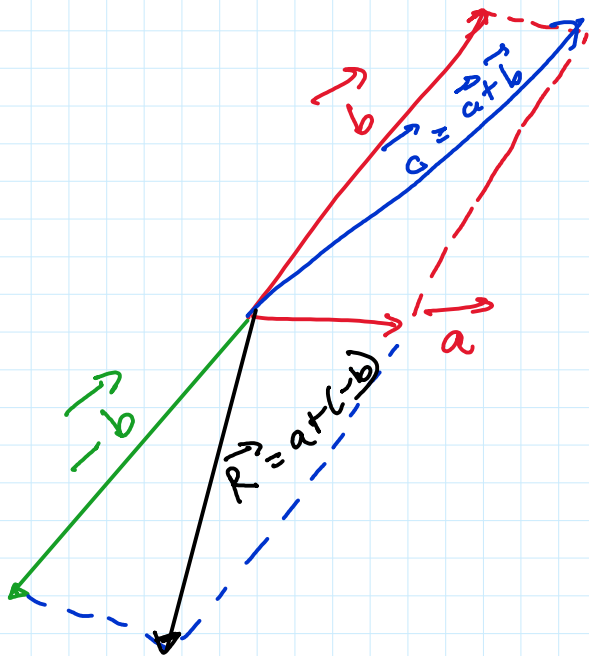
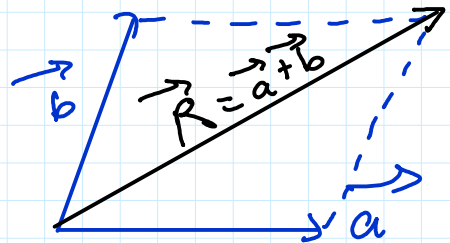
ML $\rightarrow$ minimization of (loss function)

Some Vector Laws

① (Triangle Law of addⁿ)

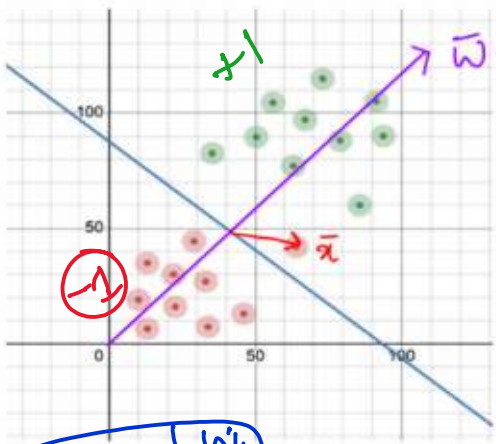


② Parallelogram Law



$$\begin{array}{c} \vec{a} + \vec{b} \\ \boxed{\vec{a} - \vec{b}} \\ \vec{a} + (-\vec{b}) \end{array}$$

Perceptron Learning Algorithm



	y_i
	+
	-

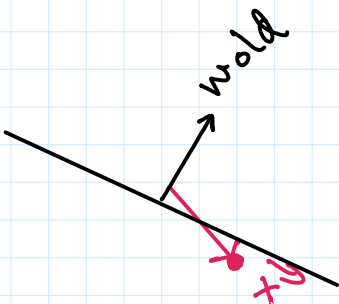
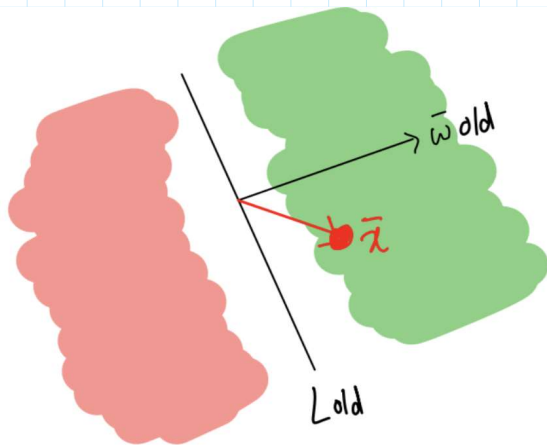
correct label = -1
Assigned label = +1

How to mathematically check if pt is misclassified?

if $y_i = \text{sign}\left(\frac{w^T x + w_0}{|w_0|}\right)$
print(correct)

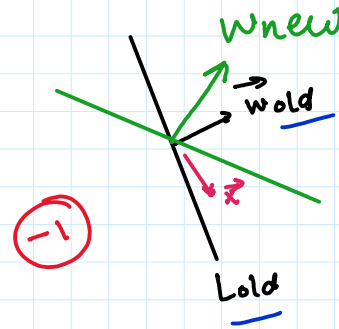
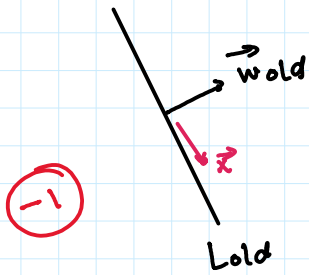
else:
print(incorrect)
→ update the classifier

(How the update happens) (How to update the classifier) (weights)
 $w^T x + w_0 = 0$

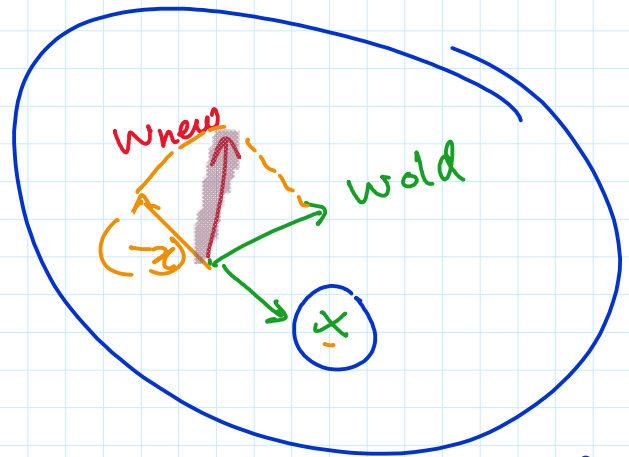


Rotate my weight vector
R

Lold



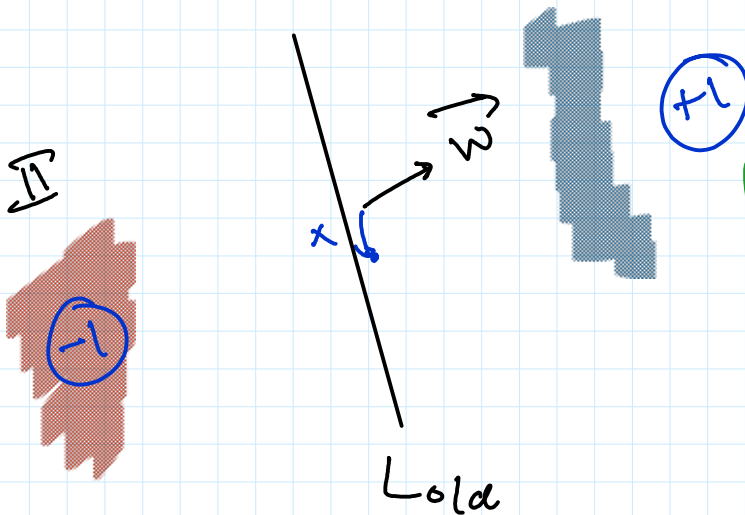
(assigned label = +1
actual = -ve)



$$w_{new} = w_{old} + (-x)$$

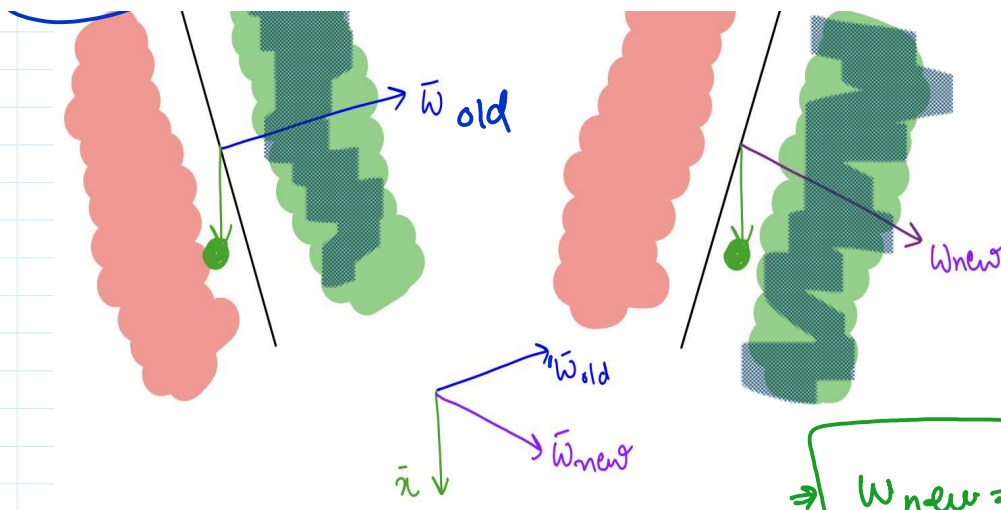
($x \rightarrow$ misclassified point)

$$w_{new} = w_{old} - x$$



assigned = -1
Actual = +1

wrong



(w_{old} and w_{new})

Summary

* (Perceptron update Rule)

① $\bar{w}_{new} = \bar{w}_{old} + \bar{x}$
 $\bar{x}$ is misclassified
 Actual $\rightarrow +1$
 Assigned $\rightarrow -1$
 Closer $y_i = +1$

② $\bar{w}_{new} = \bar{w}_{old} - \bar{x}$
 $\bar{x}$ is misclassified
 Actual $\rightarrow -1$
 Assigned $\rightarrow +1$
 Anti Closer $y_i = -1$

(one equation)

$$w_{new} = w_{old} + y_i \bar{x}$$

#

$$\vec{w}_{new} = w_{old} + y \bar{x}$$

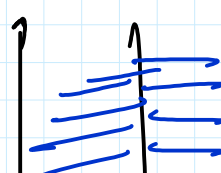
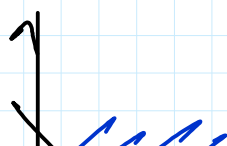
$\rightarrow$ actual label

$$y \begin{matrix} +1 \\ -1 \end{matrix}$$

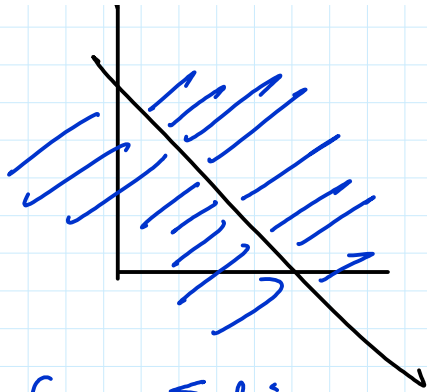
$$\begin{matrix} +1 \\ 0 \end{matrix}$$

Final thoughts =

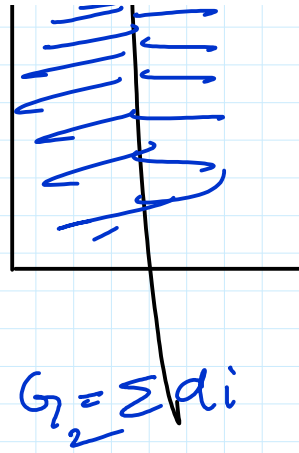
Doubts $\rightarrow$



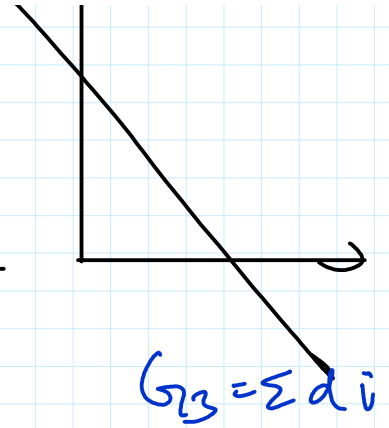
Doubts →



$$G_1 = \sum d_i$$



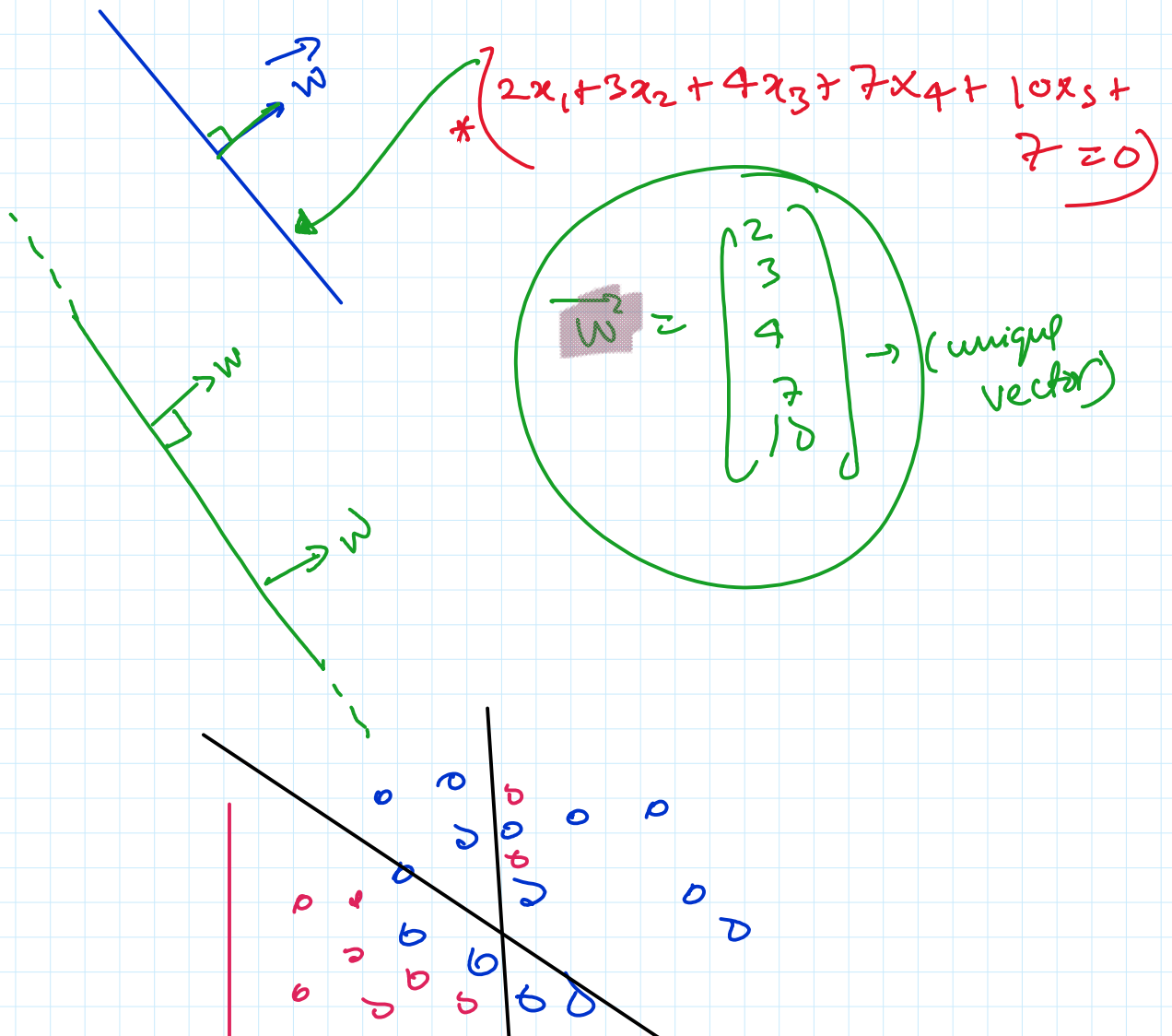
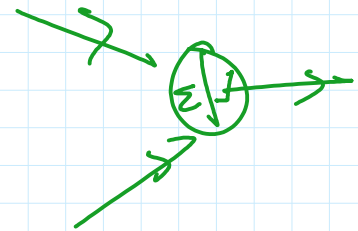
$$G_2 = \sum d_i$$



$$G_3 = \sum d_i$$

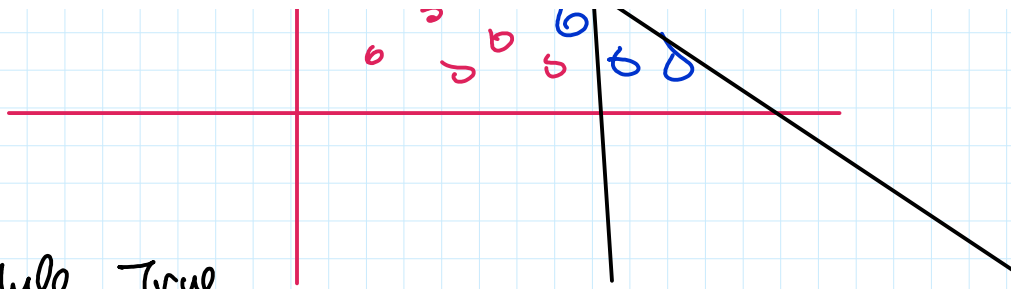
$$\max(G_1, G_2, G_3)$$

Perceptron → (Neuron Unit)



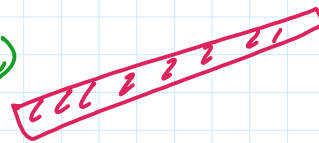
#

while True



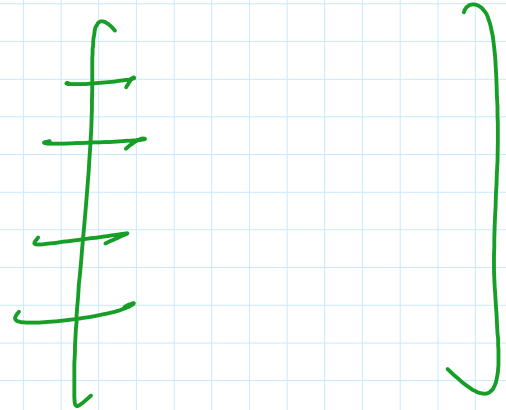
Page Rank

"Diwali" (word2vec)



[2, 3, 1, ..., 100]

WP1 = []
WP2 = []
⋮
WP-100 = []



word → to → vector

"Saurav" → [] → [2.1, 3.8, 5.7, ...]

DL-3

Doc 2 vec

